# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS TWO-CHAPTER TEST SET D

Chapters 13 & 14: **Two Variables, One Line, Math of Space: Surface Area and Volume** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The slope of the line 4x + 2y = 10 is: 1

**(a)** 1/2 **(b)** 2 **(c)** 5 **(d)** -2

**2.** Which of the following points lies on the line 3x + 2y = 12? 1

**(a)** (2, 3) **(b)** (1, 3) **(c)** (2, 2) **(d)** (0, 4)

**3.** The pair of equations 3x + 2y = 7 and 6x + 4y = 9 has: 1

**(a)** a unique solution **(b)** infinitely many solutions

**(c)** no solution **(d)** exactly two solutions

**4.** For which value of k does the pair kx + 2y = 5 and 3x + 6y = 12 have no solution? 1

**(a)** 0 **(b)** 3 **(c)** 1 **(d)** 5

**5.** The digits of a two-digit number add up to 11 and the tens digit exceeds the units digit by 3. The number is: 1

**(a)** 65 **(b)** 83 **(c)** 47 **(d)** 74

**6.** Taking $\pi = \frac{22}{7}$, the surface area of a sphere of radius $3.5$ cm is: 1

**(a)** 38.5 cm$^2$ **(b)** 308 cm$^2$ **(c)** 77 cm$^2$ **(d)** 154 cm$^2$

**7.** Taking $\pi = \frac{22}{7}$, the total surface area of a hemisphere of radius $14$ cm is: 1

**(a)** 1232 cm$^2$ **(b)** 1848 cm$^2$ **(c)** 2002 cm$^2$ **(d)** 2464 cm$^2$

**8.** The volume of a cone is _____ the volume of a cylinder with the same base and the same height. 1

**(a)** twice **(b)** one-third of **(c)** one-fourth of **(d)** half of

**9.** **Assertion (A):** A cube and a cuboid of equal volume need not have equal total surface areas. 1

**Reason (R):** Total surface area depends on the individual dimensions, not only on the product $lwh$.

**(a)** Both A and R are true and R is the correct explanation of A

**(b)** Both A and R are true but R is not the correct explanation of A

**(c)** A is true but R is false

**(d)** A is false but R is true

**10.** **Assertion (A):** The volume of a pyramid with base area $B$ and height $h$ is $\frac{1}{3}Bh$. 1

**Reason (R):** The volume of a cuboid is $lwh$.

**(a)** Both A and R are true and R is the correct explanation of A

**(b)** Both A and R are true but R is not the correct explanation of A

**(c)** A is true but R is false

**(d)** A is false but R is true

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Solve by adding and subtracting: $x + y = 12$ and $x - y = 4$. **2**

**12.** Find the points where the line $3x + 4y = 12$ cuts the axes. **2**

**13.** A tank is $2$ m long, $1$ m wide and $1.5$ m deep. How many litres of water can it hold? ($1$ m$^3 = 1000$ litres) **2**

**14.** How is a right circular cone obtained from a right-angled triangle? **2**

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** The sum of two numbers is 40, and one number is 4 more than three times the other. Find the numbers. **3**

**16.** A swimming pool is $25$ m long, $10$ m wide and $2$ m deep. How many litres of water does it hold when full? How long will a pump delivering $2500$ litres per minute take to fill it? **3**

**17.** If 1 is added to both numerator and denominator, a fraction becomes $\frac{3}{5}$; if 1 is subtracted from both, it becomes $\frac{1}{2}$. Find the fraction. **3**

**OR**

A line has slope 2 and cuts the y-axis at $(0, -3)$. Write its equation and find the point where it meets the x-axis. **3**

**18.** Find the curved surface area and the total surface area of a solid hemisphere of radius $14$ cm ($\pi = \frac{22}{7}$). **3**

**OR**

A pyramid has a triangular base with sides $6$ cm, $8$ cm and $10$ cm and a height of $9$ cm. Find its volume. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** The sum of the length and the width of a rectangle is 22 cm. If the length is decreased by 2 cm and the width is increased by 3 cm, the rectangle becomes a square. Find the original dimensions and the area of the rectangle. **5**

**OR**

The price of 8 mangoes and 5 guavas together is ₹190, and the price of 5 mangoes and 8 guavas together is ₹187. Find the price of each fruit by first adding and then subtracting the two equations. Why does this method work well here? **5**

**20.** **Case study:** A model of a pyramid has a square base of side $16$ cm and a vertical height of $15$ cm. **5**

(i) Find the slant height of each triangular face.
(ii) Find the area of the four triangular faces.
(iii) Find the total surface area including the base.
(iv) Find the volume.
(v) Why is no separate surface-area formula needed for pyramids?

**OR**

(i) Using the picture of a sphere divided into many thin cones with apex at the centre, show that $V = \frac{1}{3} \times \text{surface area} \times r$ and hence $V = \frac{4}{3}\pi r^3$. **5**
(ii) Find the volume of a sphere whose surface area is $1386$ cm$^2$ ($\pi = \frac{22}{7}$).

**END OF QUESTION PAPER** — **Total: 40 Marks**